

Space Attack

Complete Thread Transcript

115 visible messages from the project conversation
Prepared October 7, 2026 | Repository: github.com/ophiocus/space-attack

This transcript preserves the visible user and assistant messages in chronological order. Ambient browser-state notices, hidden reasoning, and tool execution logs are excluded. Non-text attachments are identified inline when present.

001. Carlos - User

get the documentation for this project here, <https://www.google.com/url?q=https://docs.google.com/forms/d/e/1FAIpQLSez2MofiQ-lcODPPXtsleaqrH0WxZ9hsuF3b0oopN3Mi1kGgA/viewform?usp%3Dsharing%26ouid%3D105971630823772469064&source=gmail&ust=1791461640875000&sa=E>;

[\[https://www.google.com/url?q=https://docs.google.com/forms/d/e/1FAIpQLSez2MofiQ-lcODPPXtsleaqrH0WxZ9hsuF3b0oopN3Mi1kGgA/viewform?usp%3Dsharing%26ouid%3D105971630823772469064&source=gmail&ust=1791461640875000&sa=E\]](https://www.google.com/url?q=https://docs.google.com/forms/d/e/1FAIpQLSez2MofiQ-lcODPPXtsleaqrH0WxZ9hsuF3b0oopN3Mi1kGgA/viewform?usp%3Dsharing%26ouid%3D105971630823772469064&source=gmail&ust=1791461640875000&sa=E)(<https://www.google.com/url?q=https://docs.google.com/forms/d/e/1FAIpQLSez2MofiQ-lcODPPXtsleaqrH0WxZ9hsuF3b0oopN3Mi1kGgA/viewform?usp%3Dsharing%26ouid%3D105971630823772469064&source=gmail&ust=1791461640875000&sa=E>)

additionally look at this email body that points to that link

Hi Carlos,

This is Keteven from G2i. Your next step is a short calibration task. It's paid at 50% of your hourly rate, capped at two hours, and included in your first payment once you join us.

Please submit it by ****tomorrow, October 8, at 9:00 PM BRT / 8:00 PM ET / 2:00 AM CEST (Oct 9th)****.

The task: build Space Attack, a small, polished browser arcade game where the player pilots a spaceship against waves of enemies.

Reference: [\[https://www.youtube.com/watch?v=jYIC8ADIArc\]](https://www.youtube.com/watch?v=jYIC8ADIArc)(<https://www.youtube.com/watch?v=jYIC8ADIArc>)

It must include:

- Keyboard movement and firing, with on-screen instructions
- Enemy waves, working collisions, a visible score and health/lives
- Increasing difficulty
- A start screen, game over and restart

Rules:

- 2 hours of active work max, including your reflection. Track your time.
- Use Codex only for all the code. No other AI tools.
- Start a fresh project and play-test it yourself.
- Save all your prompts and write the reflection in your own words.
- If something is unfinished, submit what you have and explain the issue.

Submit everything here (allow 15 to 20 minutes for the form):\

[\[https://docs.google.com/forms/d/e/1FAIpQLSez2MofiQ-lcODPPXtsleaqrH0WxZ9hsuF3b0oopN3Mi1kGgA/viewform?usp=sharing&ouid=105971630823772469064\]](https://docs.google.com/forms/d/e/1FAIpQLSez2MofiQ-lcODPPXtsleaqrH0WxZ9hsuF3b0oopN3Mi1kGgA/viewform?usp=sharing&ouid=105971630823772469064)(<https://docs.google.com/forms/d/e/1FAIpQLSez2MofiQ-lcODPPXtsleaqrH0WxZ9hsuF3b0oopN3Mi1kGgA/viewform?usp=sharing&ouid=105971630823772469064>)

If you have any questions, just reply here.

Best,

002. Codex - Assistant

I'll inspect the project files and the linked form's visible instructions, then compare those requirements with the existing project so I can surface the relevant documentation and any gaps.

003. Codex - Assistant

I opened the [G2i submission form](https://docs.google.com/forms/d/e/1FAIpQLSez2MofiQ-IcODPPXtsleaqrH0WxZ9hsuF3b0oopN3Mi1kGgA/viewform). Its instructions say to submit the Space Attack game and supporting materials, keep total time-including testing and submission prep-within two hours, and write **300-500 words across four reflection questions**. Make sure reviewers can access every link, and remove credentials or unrelated personal information from anything submitted.

The first page asks for your name, active minutes, the Codex interface and model used, and whether the game is hosted. I couldn't view later questions because the form requires those answers before it will advance, so I left them untouched. The form's title says "Solitaire - Calibration Submission," though its instructions refer to Space Attack.

The email's game requirements are: keyboard movement and firing with on-screen instructions; enemy waves and working collisions; visible score and health/lives; rising difficulty; and start, game-over, and restart screens. It also asks you to use Codex only for code, start a fresh project, play-test it, track time, save your prompts, and write the reflection in your own words. The deadline stated is **October 8 at 9:00 PM BRT / 8:00 PM ET**, which is **October 9 at 2:00 AM CEST**.

The project folder is currently empty, so there's no existing project documentation or implementation here yet.

004. Carlos - User

This is the initial contact thread, I will fork from here, rename and consider archived

005. Codex - Assistant

Understood. This thread has the initial contact and form details; you can fork it, rename the fork, and archive this one when you're ready.

006. Carlos - User

In order to complete the task we will first plan the studio that will run the project.

Plan the workflow of this gaming studio, write the results as guide files for the agents that will finish the job:

The game studio is split in departments that represent knowledge areas and you will perform the sensible roles from within these departments.

The production of the actual game will follow stages in which each department needs to contribute and clear the project stage.

The Game:

Core: this is a single player horde style game, it should comply with "twin stick shooter", with PC compatible keymaps, "top down" render composition, refreshing the ask:

- Keyboard compatible twin stick shooter layout
- Increasing difficulty and variety of obstacles, with sensible power ups, health (nice to have, ambient triggers, foe traps), high scores
- Based on the classic space attack, re imagined to take advantage of latest game technologies
- Do an online pass for free low poly 3d assets and build a local library, space sound fx
- Use the threeJS core 3D engine, create the game engine and final game around it

Nice to haves:

- The visuals should be inspired in diorama and tilt shift styles, giving the render space a cute, tiny miniature look
- Twin stick shooter should support gamepads
- Sound can be midi but Would be great to have some samples

For context remember this is meant to be a web based game to run on a browser, if you implement webGPU make sure you add a performance /CPU mode.

007. Codex - Assistant

I'll turn this into a practical studio playbook for the agents who will build the game: department responsibilities, a staged workflow with clear approval gates, and concrete technical and asset policies. I'll check the repository first, then verify current Three.js and browser-rendering guidance so the playbook stays grounded.

008. Codex - Assistant

The workspace is empty, so I'm setting up a lightweight documentation structure from scratch. The playbook will keep the two-hour cap central: small scope, explicit department sign-offs, a baseline browser renderer, and optional WebGPU only if its fallback and low-performance mode are demonstrably covered.

009. Codex - Assistant

Created the studio playbook in the fresh project:

- [AGENTS.md](/CODEX/g2i/AGENTS.md) - agent instructions and studio roles
- [studio-workflow.md](/CODEX/g2i/docs/studio-workflow.md) - production stages, department contributions, and stage gates
- [game-brief.md](/CODEX/g2i/docs/game-brief.md) - requirements and acceptance checklist
- [architecture-guide.md](/CODEX/g2i/docs/architecture-guide.md) - Three.js, input, browser, and performance guidance
- [asset-policy.md](/CODEX/g2i/docs/asset-policy.md) - initial asset research, local library conventions, and license checks
- [decisions.md](/CODEX/g2i/docs/decisions.md) - initial production and renderer decisions

The asset research points to [Kenney's Space Kit](https://kenney-assets.itch.io/space-kit), [Kenney's Simple Space](https://kenney.nl/assets/simple-space), and [Poly Haven's license page](https://polyhaven.com/license) as candidates; the guide requires checking the source license again when acquiring assets. The technical guidance reflects the current [Three.js WebGPURenderer documentation](https://threejs.org/docs/pages/WebGPURenderer.html): WebGPU is optional, with an explicit performance mode and tested fallback required if chosen.

I left the project code and assets untouched; this turn produced the workflow guides.

010. Carlos - User

about the production workflow:

Core idea: iterations, lets complete individual units of functionality starting with a solid, well principled core game engine running, from there I will steer based on iteration prompts.

Framework:

three JS core engine

js+css+html for the base program launcher (should run local with a local run warning alert bar)

your choice of framework to work the actual game engine (working title ACT engine)

godspeed

011. Codex - Assistant

I'll revise the playbook around prompt-led iterations: first deliver a complete, dependable ACT engine foundation as one reviewable unit, then build gameplay in small functional slices steered by your iteration prompts. I'll also specify the plain HTML/CSS/JS launcher and its local-run status banner.

012. Codex - Assistant

Updated the studio guides to make the workflow prompt-led and incremental.

- [studio-workflow.md](/CODEX/g2i/docs/studio-workflow.md) now defines small, complete iterations. Iteration 0 is the ACT engine foundation; after it clears, your prompts steer each next slice.
- [architecture-guide.md](/CODEX/g2i/docs/architecture-guide.md) separates the plain HTML/CSS/JS launcher from the Three.js ACT engine. I chose vanilla JavaScript ES modules as the initial engine framework.
- [AGENTS.md](/CODEX/g2i/AGENTS.md) and [game-brief.md](/CODEX/g2i/docs/game-brief.md) carry the same direction for future agents.
- [iterations.md](/CODEX/g2i/docs/iterations.md) provides a log template; it records this as planning only, with no implementation iteration completed.

The local launcher is specified to show a clear local-preview status banner and never imply the game is hosted.

013. Codex - Assistant

Iteration 2 code pass is complete and ready for your browser play-check. I did not interact with or play-test the browser.

- Separated enemy world-space movement from player transforms, with gradual steering.
- Expanded the orthographic view and sized the ship to **1/15** of view height.
- Centered loaded models; bullets now start at the ship's measured forward edge and travel along its aim direction.

- Derived collision radii from model footprints, added **F3** collision outlines, swept projectile checks, and visible hit/damage feedback.
- Added synthesized fire, hit, and damage sounds.
- Labeled the current mode **Asteroid Mode**; **Legacy Mode** appears in the menu as planned work for after the core iterations.

Checks: `npm test` passed all six focused unit tests; syntax checks and `npm run build` passed. Vite still reports its large JavaScript chunk advisory.

I recorded the changes and your prompt in [iteration-02.md](/CODEX/g2i/prompts/iteration-02.md) and [iterations.md](/CODEX/g2i/docs/iterations.md). Legacy Mode is visible but not playable yet.

014. Carlos - User

My request:

bug:

gameplay still does not make sense: wrong acceleration of enemies, damage is invisible hits are random and dont report score.

test:

remove all speed from enemies

record enemy position

move craft to maximum cardinal points

record enemy position

observe

enemy position drift

consider only root cause and fix

From here redraw the game scene so that actors are all independent (use a luna 5 agent to read from unreal actors basics online or your memory).

Then let me test again

015. Codex - Assistant

I found a likely root cause in the actor transform path: model-centering offsets were stored on the same transform that gameplay overwrites each frame. I'm checking the current patch and adding a regression test for zero-speed enemies while the player moves to each cardinal edge.

016. Codex - Assistant

The regression tests now reproduce the exact zero-speed/cardinal movement check and verify an intentionally off-center model remains centered on its own actor root through rotation and translation. Both pass. I'm recording this as a root-transform fix, then I'll run the unit suite and production build.

017. Codex - Assistant

The direct Node test runner reports all 8 tests passing. The `npm` wrappers are exiting with code 1 without diagnostics in this environment, and the build also needs to write `dist/`; I'm retrying the requested checks with the required elevated execution.

018. Codex - Assistant

Fixed the actor transform root cause. GLB normalization offsets were sharing the transform gameplay overwrites each frame. Now each actor has its own world-transform root, with the centered model kept on a separate visual child.

Added regression checks for a zero-speed enemy while the player moves to all four cardinal extremes, and for an offset model rotating and translating without its visual center drifting. **All 8 unit tests pass; the production build succeeds.** Vite still reports its large-chunk advisory.

I didn't browser-test; the game is ready for your play check. [engine.js](/CODEX/g2i/src/act/engine.js) ? [simulation.test.js](/CODEX/g2i/tests/simulation.test.js) ? [iteration notes](/CODEX/g2i/docs/iterations.md)

019. Carlos - User

My request:

CRITICAL GAMEPLAY PASS WELL DONE

Minor: the ship is backwards on the screen up down axis fix only that and tell me to check again.

020. Codex - Assistant

I'll trace the ship's model-forward direction against the game's up-screen aim vector, adjust only that orientation mapping, and run the focused unit/build checks.

021. Codex - Assistant

The bundled craft's nose points along local -Z, while the existing aim-to-yaw mapping assumed +Z. I've corrected the player's yaw mapping only, keeping the aim vector and projectile path unchanged; I'm checking all four cardinal directions in the unit test.

022. Codex - Assistant

Fixed the ship's reversed up/down orientation by mapping aim to the bundled craft's local nose direction. Aim and firing are unchanged.

All 9 unit tests pass, and the production build succeeds. **Please check again in the game tab.**

023. Carlos - User

My request:

Consider the core engine complete.

we will work on the modes, we have two modes, one is engine complete (asteroid), we now need to create the legacy mode for proper space invaders.

roadmap

- L0: learn the space invaders legacy rules: enemies zig zag downwards towards a player that can only move on the left right dimension on the ground, firing up. Zone of the enemies fire slow rounds at sparse intervals. The terrain has floating blockers that get damaged both ways in cross fire, stopping rounds
- L1: do an architectural plan: two game modes on the same engine, look for code dependencies in the asteroid mode that would need to be moved out or elsewhere so as to make the game modes fully modular
- L2: Stub the new legacy game mode and implement
- L3: Hand over to test

For simplicity current advanced assets (ship that can fire enemies that move) can be re used then reworked in the new game mode, either override or clone, whatever is simpler.

024. Codex - Assistant

I'll follow L0-L3 in order: verify the classic movement/fire/blocker loop, map mode boundaries onto the current ACT engine, then implement Legacy as an isolated mode and hand it over with unit checks. I'll leave browser play-testing to you.

025. Codex - Assistant

L0 confirms the classic loop: the cannon moves only horizontally and fires upward; the formation sweeps side to side, steps downward on each edge reversal, and becomes more urgent as it thins. Enemy shots descend, shots stop on damaged shields, and invasion at the bottom ends the run. I'm keeping this mode's formation, firing, shields, and scoring out of the Asteroid simulation while sharing the ACT renderer, local assets, input shell, and score/lives HUD.

026. Codex - Assistant

The Legacy rules and controller are in place. The first unit run caught a mismatch in my invasion-threshold test fixture (the implementation's cutoff is player position minus the configured margin); I'm correcting the fixture and then checking the build and both-mode lifecycle wiring.

027. Codex - Assistant

The first complete Legacy slice now has independent state and a `start/update/dispose` lifecycle; the shared shell dispatches exactly one mode at a time and hides the Asteroid arena while Legacy runs. Unit coverage now checks edge descent, thinning/wave speed-up, invasion, row scoring, and shield damage from upward and downward shots. I'm checking the launcher's wiring and production bundle next.

028. Codex - Assistant

The production bundle builds and all 14 unit tests pass, including a controller lifecycle test that starts Legacy, moves the cannon, fires, and verifies cleanup. Manual play is still pending for your hands-on pass. One architecture note is explicit in the plan: Asteroid's older implementation is still in `main.js`; its extraction is staged after Legacy's play check to avoid a wide refactor during this feature slice.

029. Codex - Assistant

Legacy Mode is available from the landing menu. It has horizontal-only cannon movement (`A/D` or `<-/->`), one upward shot at a time, a five-row formation that reverses and descends, sparse enemy fire, and destructible shields made from the locally bundled Kenney platform asset. Clearing formations increases the pressure; enemy rows score differently.

****L3: please test Legacy Mode in the open game tab.**** Try movement and firing, both sides' shots hitting shields, formation descent, score and wave changes, invasion/game over, pause/restart, then switch back and confirm Asteroid Mode still plays.

All 14 unit tests pass, and the production build succeeds. I did not browser-test, per your instruction. The architecture plan records one remaining boundary: Asteroid's implementation is still in `main.js` behind a mode lifecycle adapter; extracting it into its own module is deferred until after your Legacy play check.

The rule summary is documented in [mode-architecture.md](/CODEX/g2i/docs/mode-architecture.md), based on the [Taito rules](https://taito.co.jp/en/steam/sie/help) and [Atari manual](https://www.atari.compendium.com/archives/manuals/7800/space_invaders.pdf). Code and test notes are in [legacy.js](/CODEX/g2i/src/modes/legacy.js), [legacy-simulation.js](/CODEX/g2i/src/modes/legacy-simulation.js), and [iteration log](/CODEX/g2i/docs/iterations.md).

030. Carlos - User

My request:
legacy mode:

Bug: the haste is like, cube of haste, when it should be haste plus some haste number minus remaining enemies or like that. calculate that the last enemy's speed cannot be more than thrice the ship's speed.

Both modes: Enemies primer

Legacy: the invaders all have the same speed, ai attributes, but are different shapes.

Asteroid: in the original asteroid the (enemy)rocks would split on impact, one type splits into other two types, and so on like a binary tree.

Gameplay changes:

LEGACY: look for more alien looking shapes in the assets and generate once a randomized grid for the spawns, change the grid by an offset in a closed array of enemy model references to give the enemy grid processor cheap randomness (the offset is the only runtime random)

ASTEROID: use the same random closed loop offset arrangement to order the breakdown on impact of enemy ships, Enemies are now three types, A,B,C, a splits into two b splits into two c they spawn in waves according to horde game logic

031. Codex - Assistant

I'll fix the Legacy speed curve first, then inspect the bundled asset library for distinct alien forms and implement the shared enemy primer: randomized Legacy formation layouts and A->B->C asteroid splitting with reproducible closed-loop ordering. I'll keep mode rules separate and verify with unit tests/build only.

032. Codex - Assistant

I'm treating this as the next gameplay iteration. I'll preserve the Legacy speed cap and deterministic formation rule, add the A->B->C split loop to Asteroid mode, then run the project's unit tests and build; I'll leave browser play-testing to you.

033. Codex - Assistant

The two requested rules are now in code: Legacy haste is computed from the fixed base speed and defeated count with a hard 25.5 cap, while invader visuals cycle from one wave-level random offset and share movement/fire behavior. Asteroid roots now split into smaller B then C enemies using an eight-direction closed loop. I'm checking for integration mistakes and will record the assets and iteration outcomes.

034. Codex - Assistant

Unit checks caught an incorrect expectation in the new direction-loop test, not a runtime defect; I've corrected the expected wrapped direction. The production build command also exited without a Vite diagnostic in the sandbox, so I'm retrying the same build with the approved build-tool permission.

035. Codex - Assistant

The final review confirms Legacy has only one random draw for enemy-formation model assignment; its enemy shooter and firing cadence are deterministic. Asteroid fragmentation gets one starting offset per run, then follows the loop without additional randomness. I'm rerunning the final checks after a small initialization cleanup.

036. Codex - Assistant

Implemented the Legacy and Asteroid enemy changes.

- **Legacy:** Haste now increases additively from a fixed base, with speed capped at three times the cannon's speed. Each formation cycles through three locally bundled silhouettes; invaders share the same speed and formation behavior.
- **Asteroid:** A enemies split into two B enemies, and B enemies split into two C enemies. Split directions follow a closed loop with one randomized starting offset per run.

All 16 unit tests and syntax checks pass. The production build succeeds, with Vite's existing large-bundle advisory. I did not browser play-test; both modes are ready for your check.

The exact prompt and iteration notes are in [iteration-06.md](/1:/CODEX/g2i/prompts/iteration-06.md) and [iterations.md](/1:/CODEX/g2i/docs/iterations.md).

037. Carlos - User

My request:

The game modes are technically complete.

tune-ups:

asteroid:

Replace the use of speeder* models, replace them with rock crystal large and rock large models.

The actual enemies are composit models: take the same model twice, flip one horizontally so their bases overlap

Enemies no longer shoot at player if they ever did.

Increase the camera distance from the calculated 1/15 to 1/21

show me

038. Codex - Assistant

I'll update Asteroid enemy visuals, remove the enemy fire behavior if present, and adjust the ship's view-relative scale from 1/15 to 1/21. Then I'll run unit checks and rebuild; I'll leave browser play-testing to you.

039. Codex - Assistant

Asteroid enemies now use the large crystal/rock assets as mirrored composites: each actor contains two copies centered together, with one flipped horizontally. I found no Asteroid enemy projectile or firing code; its damage remains contact-based. I've also changed the shared view-scale helper to 1/21 and am updating its unit coverage and asset records.

040. Carlos - User

My request:

Last tune up, for both modes, visuals

It is now time to fully focus on diorama camera works and tilt shift visuals

confirm you can add gpu support with the cpu cutover or confirm working platform scope at delivery

Lastly prepare a thank you scroller that starts after game over screen:

lists all assets and frameworks used, with urls where viable

lists all the people involved in the process (ask me for names later add placeholders)

show my name as author, yours as co author (yes, i respect you that way)

ends with a generic ginger bread man recipe

041. Codex - Assistant

I'll fold this into the current visual tune-up. I'll keep Three.js WebGL as the delivery target, add a low-cost switch that disables tilt-shift and reduces render resolution, and label it as a performance mode rather than claiming a CPU renderer fallback. I'll also add the post-game-over credits scroller with Carlos as author, me as co-author, placeholders for other participants, asset/framework links, and the recipe ending.

042. Carlos - User

My request:

Cast:

Carlos: Caffe ingestion

Sandra: Homemaker

Antonia: Morale Booster

043. Codex - Assistant

Got it. I'll put those names and roles into the credits, keep placeholders for any remaining contributors, and preserve this cast note with the iteration record.

044. Codex - Assistant

The credits now include Carlos, Sandra, and Antonia with the roles you gave, while retaining one placeholder for additional contributors. I've also made the composite collision radius account for the doubled width. I'm doing the final unit/syntax check and build now; the renderer scope is WebGL2/GPU-accelerated desktop browsers, with a reduced-cost mode rather than a separate CPU software renderer.

045. Codex - Assistant

Ready for you to inspect in the local preview.

Both modes now get a diorama-style focus band and subtle vignette. Asteroid enemies use mirrored pairs of large crystal/rock models, and the ship scale uses the 1/21 view ratio. The game-over screen has a **Thank You** action that starts the credits scroller, with your cast list, author/co-author credits, asset and framework links, remaining-name placeholder, and gingerbread recipe.

Platform scope: desktop browsers using Three.js WebGL2 with GPU acceleration. Performance mode lowers resolution and bypasses post-processing; there is no separate CPU software-renderer fallback.

All 17 unit tests pass, syntax checks pass, and the production build succeeds. I didn't run a browser play-test, so please check the visual effect and credits in the preview.

Details: [iteration log](/I:/CODEX/g2i/docs/iterations.md) ? [asset manifest](/I:/CODEX/g2i/docs/asset-manifest.md) ? [time log](/I:/CODEX/g2i/docs/time-log.md)

046. Carlos - User

My request:

game mode tune up:

asteroid:

Rocks are not sentient, lol:

- rocks spawn at a fixed velocity, broken up rock spawns accelerated in random at a new random vector
- rock space logic folds onto the screen: a rock that exits the screen in an angle enters at the other side of the screen in the same vector

legacy:

- make ships rate of fire constant, the rate should be a function of average time to success under a given hit rate, to normal mode is all enemies killed at 60% accuracy
- when an invader touches a barrier it crumbles to the ground with no hit box

047. Codex - Assistant

I've removed player-seeking motion from Asteroid rocks: root rocks travel inward at a fixed speed, fragments get a faster random vector, and edge exits wrap to the opposite side without changing velocity. Legacy now fires on a fixed cadence based on a 33-second reference clear at 60% accuracy; shield contact removes invaders from active collision immediately while a short crumble animation plays. Unit tests cover those behaviors.

048. Carlos - User

My request:

[face with tears of joy] bro: make

- the barrier

crumble to the ground

not

- the invader

too funny, cant you see that breaks the lose state?

049. Codex - Assistant

You're right-I interpreted that backwards, and removing the invader would undermine the invasion loss condition. I'll change the contact response so the barrier sinks and loses collision, while the invader stays active and continues toward the cannon. I'll also add a regression check for that exact sequence.

050. Codex - Assistant

Fixed-the **barrier** crumbles and drops out of collision when an invader touches it. The invader stays active and keeps advancing, so it can still reach the cannon line and trigger the loss state.

Asteroid rocks now keep fixed inward velocity, wrap across screen edges without changing direction, and split into fragments with faster random vectors. Legacy fire uses a constant cadence calibrated for a 55-invader clear at 60% accuracy.

All 20 unit tests pass; syntax checks and the production build pass. I did not browser play-test. The 33-second clear target is a tuning assumption for you to judge in play.

Updated [Legacy mode code](/!:/CODEX/g2i/src/modes/legacy.js), [Asteroid motion code](/!:/CODEX/g2i/src/modes/asteroid-simulation.js), and [iteration notes](/!:/CODEX/g2i/docs/iterations.md).

051. Carlos - User

My request:

legacy

increase rate of fire by 20% success accuracy required to 40%

052. Codex - Assistant

I'll apply the requested 20% fire-rate increase and update the Legacy tuning target to 40% accuracy. At that cadence, a 55-invader wave projects to about 41 seconds of continuous fire at 40% hits; I'll verify that calculation and rerun the unit checks/build.

053. Codex - Assistant

The cadence is now 20% faster than the previous setting. The 40% accuracy model projects 138 shots for 55 invaders, or about 41.4 seconds of continuous fire. All 20 unit tests pass; I'm confirming the final build.

054. Codex - Assistant

Legacy fire rate is **20% faster** than the previous setting. The 40% accuracy model projects 138 shots-and about 41.4 seconds of continuous fire-to clear 55 invaders.

All 20 unit tests pass, and the production build succeeds. I didn't browser play-test.

055. Carlos - User

My request:

last:

visual pass

tilt shift focus should always be the ship, seen clear, everything around slowly defocusses with distance

find a camera angle that is not zero on any axis, and still frames the gameplay correctly, so no exaggerated tilt on any axis

asteroid: give the props in the scene some space bound in place rotation , vary their sizes, they are background only and have no impact only visual filler

invaders, try to find a camera tilt that would allow use to place some decorative props on the floor

056. Codex - Assistant

I'll make the focus plane follow the player so the ship stays sharp while blur increases smoothly with distance. I'll use a slight oblique camera with nonzero X/Y/Z offsets, then vary and gently rotate the Asteroid background props without adding collisions or gameplay effects. For Legacy, I'll check whether a small floor prop can be placed without obscuring the invader field or shields.

057. Codex - Assistant

The camera and focus changes are in. Asteroid scene dressing is now visual-only, with no hitboxes or collision entries; I've also placed two small non-colliding Legacy floor props beyond the main formation. I'm checking unit coverage and the production build now.

058. Codex - Assistant

The visual pass is ready for your browser check.

- The camera now uses a mild oblique angle, and tilt-shift focus follows the player ship.
- Asteroid props vary in size and rotate slowly as background decoration; they have no hitboxes or gameplay impact.
- Legacy has small decorative props at the far edge of the shared floor.

All 20 unit tests pass, and the production build succeeds. I did not play-test in the browser, so please check the framing and blur strength in your open game. The build still reports its existing large-chunk advisory.

I recorded the prompt and iteration in [iteration-11.md](/CODEX/g2i/prompts/iteration-11.md) and [iterations.md](/CODEX/g2i/docs/iterations.md).

059. Carlos - User

My request:

invaders: cap the dissolve, its too strong where the invaders are, also reduce they overall haste at the end to 2x only

060. Codex - Assistant

I'll soften the tilt-shift around the Legacy formation while keeping the ship focused, then cap late-wave invader speed at twice the player's speed. I'll check the existing haste tests and update the iteration notes.

061. Codex - Assistant

Reduced the maximum tilt-shift blur and capped Legacy invader haste at **2x cannon speed**. All 20 unit tests pass, and the production build succeeds.

The updated scene is ready for your visual check in the browser.

062. Carlos - User

My request:
invaders increase rate of fire 30%

063. Codex - Assistant

I'll increase the Legacy firing rate by 30% from its current setting, update the cadence test and documentation, then run the unit tests and production build.

064. Codex - Assistant

Legacy invaders now fire **30% faster** than the previous setting. All 20 unit tests pass, and the production build succeeds. The faster cadence models a roughly 31.8-second continuous-fire clear for a 55-invader wave at 40% accuracy.

065. Carlos - User

My request:
[ok hand sign]prepare the form, read it and tell me how it expects delivery

Did you add a thank you for the assets with a link?

066. Codex - Assistant

I'll read the Google Form's questions and compare its requested deliverables with the project's credits screen, then summarize how to submit and whether the asset thanks include a link.

067. Codex - Assistant

I opened the form and read its first section. It's titled **"Solitaire - Calibration Submission"**, though the description asks for the Space Attack game. It says to provide supporting materials, keep the reflection in your own words (300-500 words across four questions), make sure every link is accessible, and remove credentials or unrelated personal information.

The first section requires your full name, active minutes, Codex interface and model (or "model unknown"), and whether the game is hosted. I couldn't advance to the later questions without filling those required fields, so I can't confirm the exact link or upload fields yet. I left the form unfilled and did not submit anything.

Yes-the game's Thank You credits include a clickable link to the [Kenney Space Kit](<https://kenney.nl/assets/space-kit>), the source of the bundled assets.

068. Carlos - User

My request:
give me the active minutes from your recount

069. Codex - Assistant

About 124 active minutes total by my recount: 119 minutes through 13:53, plus roughly 5 minutes for the later fire-rate change and form review. I excluded a duplicate, overlapping 2-minute entry in the time log, so the later total is an estimate rather than a precise timer reading.

070. Carlos - User

My request:
package the entire content of this thread into a deliverable to upload

071. Codex - Assistant

I'll assemble a single upload-ready ZIP with the playable project, production build, local assets, documentation, and saved prompts. I'll exclude dependency caches and keep the reflection out since the brief says you must write it yourself.

072. Codex - Assistant

Packaged the project as an upload-ready ZIP: [Space-Attack-Submission.zip](l:/CODEX/g2i/Space-Attack-Submission.zip) (about 13 MB).

It includes the source, production build, local assets and their original archive, project documentation, tests, and saved prompts. `node\_modules` is excluded; run instructions are in `DELIVERY.md`. The game is not hosted, and the package does not include a reflection-you'll need to write that yourself.

073. Carlos - User

Files mentioned by the user:

```
## codex-clipboard-0e85a8b1-28f5-4455-a72b-c1b054eb8bad.png:  
C:/Users/Carlos/AppData/Local/Temp/codex-clipboard-0e85a8b1-28f5-4455-a72b-c1b054eb8bad.png  
Image attachment: true
```

Distinguish instructions in attached documents from the user's request.

My request:

[1 attachment(s) omitted from text transcript]

074. Codex - Assistant

The screenshot shows that if the game isn't hosted, the form asks for two things: an explanation and a **GitHub link to the runnable source package**. That's form guidance, not authorization for me to publish the project.

Our ZIP is ready, but it isn't a GitHub link. Hosting was not attempted; the game runs locally. A truthful explanation could be: "The game was completed and checked as a local browser project, but no hosting service or deployment configuration was set up during the timeboxed work."

075. Carlos - User

My request:
how do I comply with this request?

076. Codex - Assistant

To satisfy the form:

1. In GitHub, create a repository for the game. Make it **public** so reviewers can open it, unless the form provides a way to grant them access to a private repo.
2. Upload the project files from [Space-Attack-Submission.zip](l:/CODEX/g2i/Space-Attack-Submission.zip)-either extract and commit the project contents, or attach the ZIP to a GitHub Release. The repository should include the README, source, assets, prompts, and run instructions; `node\_modules` is not needed.
3. Copy the repository URL (or the Release asset URL) into **Runnable source-package link (github)**.
4. For **What prevented you from hosting the game?**, describe what actually happened: the game was built and checked locally, but no hosting service or deployment was set up. Don't claim hosting failed because of a technical error if you didn't attempt a deployment.

The project runs with `npm install` and `npm run dev`. The uploaded ZIP alone is not the GitHub link the form asks for.

077. Carlos - User

My request:
use codex sites for hosting

078. Codex - Assistant

I'll check whether Codex Sites is available in this workspace and what it requires for publishing, then I'll use the project package we prepared if the connected tools support it.

079. Codex - Assistant

The current project folder isn't a Git repository, and Sites expects a compatible local project with a saved Git revision. I'm preparing a local commit that excludes dependency caches and the ZIP itself; the Sites publishing controls aren't exposed to this session, so I'll report exactly where the handoff is needed.

080. Codex - Assistant

I prepared the project for Sites as a local Git repository and saved commit `adef5eb`. It hasn't been uploaded or published.

This session doesn't expose the Codex Sites controls, and the browser tab I could access is signed out, so I can't complete the deployment here. Open **Sites** in the ChatGPT desktop app, start from the local project at `I:\CODEX\g2i`, and ask:

> Deploy this project with Sites. Check compatibility, make any required changes, save a version, and give me the deployment URL.

Review the preview, then publish it with access that lets the G2i reviewers open it. Sites creates a production URL when you deploy a saved version; public publishing depends on your account and workspace settings. [Codex Sites guide](https://developers.openai.com/codex/sites), [Sites publishing help](https://help.openai.com/en/articles/20001339-creating-and-using-chatgpt-sites).

081. Carlos - User

My request:
show me how to step by step

082. Codex - Assistant

Deploy Space Attack with Codex Sites

1. Open the **ChatGPT desktop app** and select **Codex**.
2. Open **Sites** from the sidebar. Choose the option to start from an **existing local project**, then select `I:\CODEX\g2i`. The project is prepared as a Git repository at commit `adef5eb`.
3. Ask Sites:

> Deploy this project with Sites. Check compatibility and make any required changes. Save a version for review, but don't deploy it yet.

4. Review the preview. Check that the game loads, both modes are selectable, and the local `.glb` assets load correctly. Ask Sites to fix any issues you find.

5. When the preview is ready, ask:

> Deploy the reviewed version and give me the URL.

Sites saves a version first, then deploys it. Each deployment URL is live, so review the preview before this step. ([Codex Sites guide](https://developers.openai.com/codex/sites))

6. In the sharing settings, choose **Anyone on the internet** if available, then publish and copy the URL. Public access depends on your account and workspace permissions; if the option is missing, an admin may need to enable it. ([Sites publishing help](https://help.openai.com/en/articles/20001339-creating-and-using-chatgpt-sites))
7. Open the deployed URL in Chrome and check it once more. Then use that URL in the form's hosted-game field.

If Sites can't deploy the existing project, keep the ZIP package as the runnable source deliverable and describe the specific compatibility or access issue in the form.

083. Carlos - User

My request:
there is no sites, confirm free tier can use sites

084. Codex - Assistant

Confirmed: ChatGPT Free does not include Sites. OpenAI's release notes say Sites is unavailable on Free and Go; it's available on eligible paid plans, subject to rollout, region, and workspace settings. That's why you don't see it. [OpenAI release notes](https://help.openai.com/en/articles/6825453-chatgpt-release-notes)

I gave you the wrong impression earlier when I described the deployment steps as if Sites were available in your account.

085. Carlos - User

My request:

so the option is to host it myself in git, how to host it so its playable?

086. Codex - Assistant

I'll check the project's asset paths and the current GitHub Pages guidance so I can give you steps that will work for this Vite game, not just a generic Git upload.

087. Codex - Assistant

I found one GitHub Pages issue in this game: its model loader uses root-absolute asset paths, which break when Pages serves the app under a repository subpath. I'll make the Vite base path and model URLs work for that layout and add a GitHub Actions Pages workflow, so after you create the public repository and push, GitHub can build and publish it.

088. Codex - Assistant

GitHub Pages will work for this Vite game. I've prepared the project for it: the asset paths handle GitHub's repository subpath, and a GitHub Actions workflow now runs the tests, builds the game, and deploys it. I also refreshed the [upload ZIP](/CODEX/g2i/Space-Attack-Submission.zip). The latest local commit is `f4141e2`; it has not been pushed or published.

1. Create a **public** repository on GitHub, for example `space-attack`. Leave "Add a README" unchecked so the new repo starts empty. GitHub Free supports Pages for public repositories. [GitHub Pages docs](https://docs.github.com/en/pages/getting-started-with-github-pages/using-custom-workflows-with-github-pages)
2. In the project terminal, connect that repository and push the `main` branch. Replace `` with your GitHub username:

```
``powershell
git remote add origin https://github.com/<YOUR_USERNAME>/space-attack.git
git push -u origin main
``
```

3. In the repository on GitHub, open **Settings -> Pages** and set **Source** to **GitHub Actions**.
4. Open **Actions** and wait for **Deploy Space Attack to GitHub Pages** to finish. It runs the unit tests and build before publishing.
5. Open the URL shown by the successful deployment, usually `https://<YOUR\_USERNAME>.github.io/space-attack/`, and put it in the form's hosted-game field. The Vite base path is configured for this URL format. [Vite deployment guide](https://vite.dev/guide/static-deploy)

The site and repository will be public. The 20 unit tests pass, and the build was checked with the GitHub Pages subpath. I haven't pushed or browser-tested the live deployment.

089. Carlos - User

My request:

I have connected you to github, can you create the new remote?

090. Codex - Assistant

I'll check the connected GitHub capabilities, then create a public `space-attack` repository and push the prepared `main` branch if the connection supports repository creation.

091. Codex - Assistant

The available GitHub CLI credential is present but invalid, and I don't see a GitHub repository connector in this session. I'm checking whether the browser has an authenticated GitHub session before I tell you what's blocking repository creation.

092. Codex - Assistant

I've prepared the local branch. I'm checking the current GitHub connection and repository state once more; if authentication is still unavailable, I'll give you the exact one-step unblock.

093. Codex - Assistant

I can create it, but the GitHub connection isn't authenticated in this session: the saved `gh` token is invalid, and there's no remote configured yet. The local `main` branch is clean and ready.

Please re-authenticate with `gh auth login -h github.com`, then tell me when it's done. I can then create `ophiocus/space-attack`, add it as `origin`, and push `main`. The repository would be public for the GitHub Pages hosting plan.

094. Carlos - User

My request:
re install the plugin?

095. Codex - Assistant

I'll check the available plugin-management options. The failure I saw was the GitHub CLI credential, so reinstalling may not be the fix.

096. Codex - Assistant

No-the GitHub plugin is already installed and enabled. The problem is the separate GitHub CLI login: its saved token is invalid. Reinstalling the plugin is unlikely to fix that; re-authenticate with `gh auth login -h github.com`, then I can create the remote and push the branch.

097. Carlos - User

My request:
in what terminal?

098. Codex - Assistant

Use a PowerShell terminal opened in the project folder `I:\CODEX\lg2i`-the Codex terminal or Windows Terminal is fine. Run:

```
```powershell
gh auth login -h github.com
```
```

Follow the prompts to sign in through your browser. When it confirms you're logged in, tell me and I'll create the remote.

099. Carlos - User

My request:
no you got it wrong, use powershell to figure out the current ssh configurations

100. Codex - Assistant

Understood-you mean use the machine's existing SSH setup. I'll inspect the SSH config, loaded key fingerprints, and Git remotes in PowerShell, without opening or printing any private key contents.

101. Codex - Assistant

You're right; the SSH config was the relevant thing to check. In PowerShell, I confirmed GitHub accepts the configured key: it maps `github.com` to `~/ssh/id\_ophiocus`, and GitHub authenticated it as **ophiocus**.

There's no remote configured yet. SSH can push to a repository, but it can't create the repository itself. Please create an empty `space-attack` repository on GitHub (public, for the Pages plan), then I can add it as `origin` and push `main`.

102. Carlos - User

My request:
remote is up

103. Codex - Assistant

Great. I'll verify the expected `ophiocus/space-attack` SSH remote is reachable, then add it as `origin` and push the prepared `main` branch.

104. Codex - Assistant

The repository is reachable and the SSH key is accepted. The sandbox blocked writing `.git/config`, so I'm requesting permission to add the `origin` entry; after that I'll push `main`.

105. Codex - Assistant

Done. I added `origin` as `git@github.com:ophiocus/space-attack.git` and pushed `main`. The branch now tracks `origin/main`, and the working tree is clean.

Repository: github.com/ophiocus/space-attack

106. Carlos - User

My request:

where is the runnable link of the built thing?

Also: compile this entire thread into a single PDF and add to this very repo

107. Codex - Assistant

I'll check the Pages deployment to find the actual play URL. I'll also pull the thread record and use the PDF workflow to create a transcript artifact, then add it to the repository.

108. Codex - Assistant

I'm applying the PDF skill's document workflow for the transcript file. First I'll inspect the deployment status and locate the full thread history so the PDF reflects the conversation rather than a summary.

109. Carlos - User

My request:

it didnt run

Annotations

2 errors, 1 warning, and 1 notice

```
| [**build**](https://github.com/ophiocus/space-attack/actions/runs/37675592606/job/112978141050#step:7:7) | HttpError: Not Found - [https://docs.github.com/rest/pages/pages#get-a-apiname-pages-site](https://docs.github.com/rest/pages/pages#get-a-apiname-pages-site)
|
-----
-----
-----
[**HttpError:&#x20;**](https://github.com/ophiocus/space-attack/commit/f4141e23ada5b9c6b9e67a31a7b5a50fa9ff9650#annotation_86067479164)[/home/runner/work/_actions/actions/configure-pages/v5/node_modules/@octokit/request/dist-node/index.js#L124](https://github.com/ophiocus/space-attack/commit/f4141e23ada5b9c6b9e67a31a7b5a50fa9ff9650#annotation_86067479164)
[**build**](https://github.com/ophiocus/space-attack/actions/runs/37675592606/job/112978141050#step:17:2)
[**build**](https://github.com/ophiocus/space-attack/actions/runs/37675592606/job/112978141050)
```

110. Codex - Assistant

I found the deployment failure: the workflow can build, but GitHub Pages has not been enabled for this repository, so `configure-pages` returns 404. I'll update the workflow to enable Pages during deployment, then rerun it and use the published URL if it succeeds.

111. Carlos - User

My request:

I think I dont have more pages to setup, It would undo another page?

112. Codex - Assistant

No. Enabling Pages for this repository creates its project site at `https://ophiocus.github.io/space-attack/`. Your account-level site at `https://ophiocus.github.io/` and Pages sites for other repositories keep their own settings. I'm checking the official URL rules while I finish the PDF.

113. Codex - Assistant

The thread export contains 112 messages from this task, including the original brief, iterations, and hosting discussion. I'm creating a text transcript PDF and will add it under `output/pdf/`; the current Pages build is still blocked on the repository setting.

114. Carlos - User

My request:

https://ophiocus.github.io/space-attack/ when?

115. Codex - Assistant

It goes live after Pages is enabled for `space-attack` and its Actions deployment succeeds. The current run failed before publishing; the project URL is separate from your account site and other repos. The browser session is signed out, so I can't change that repo setting from here.